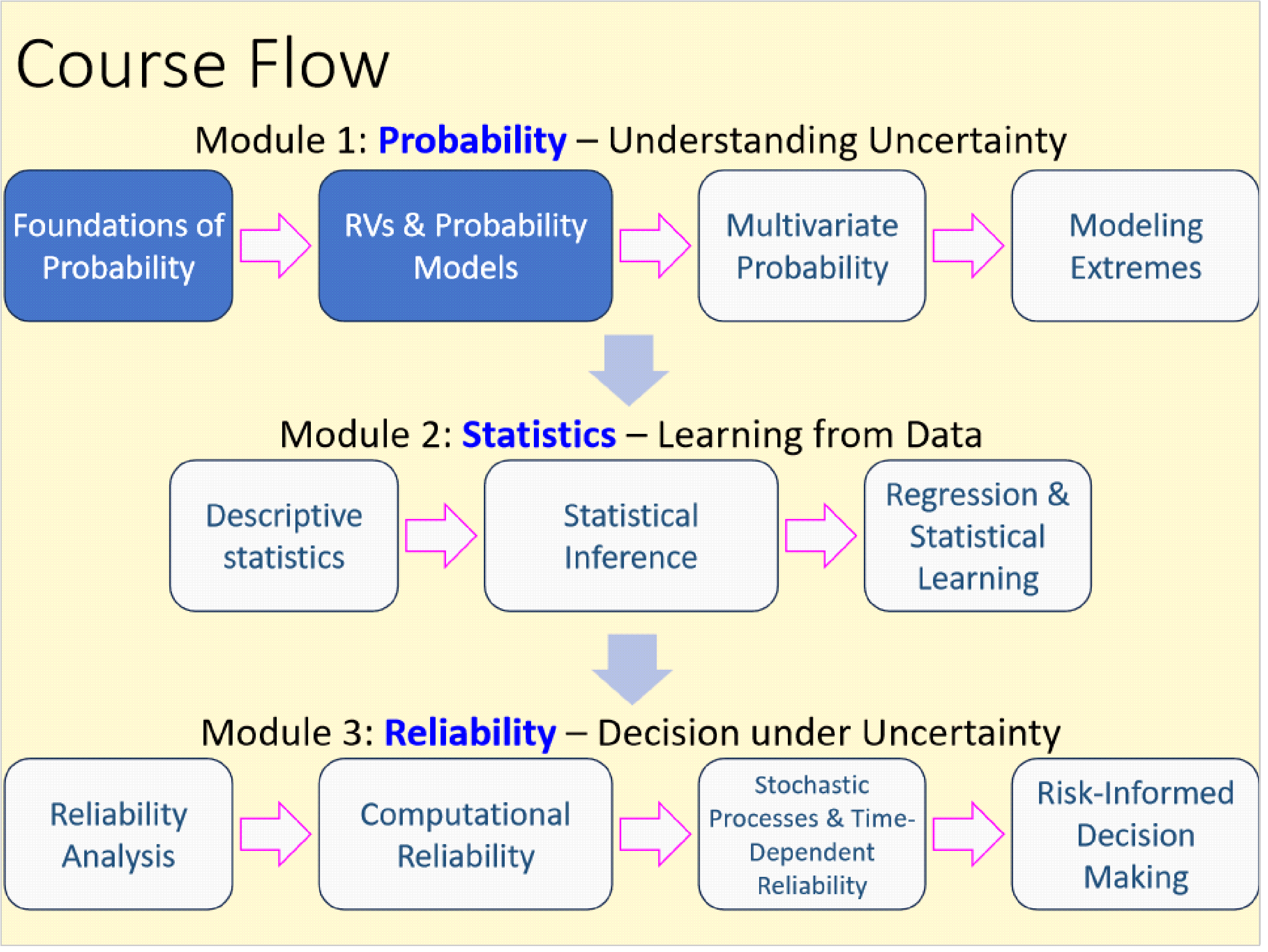
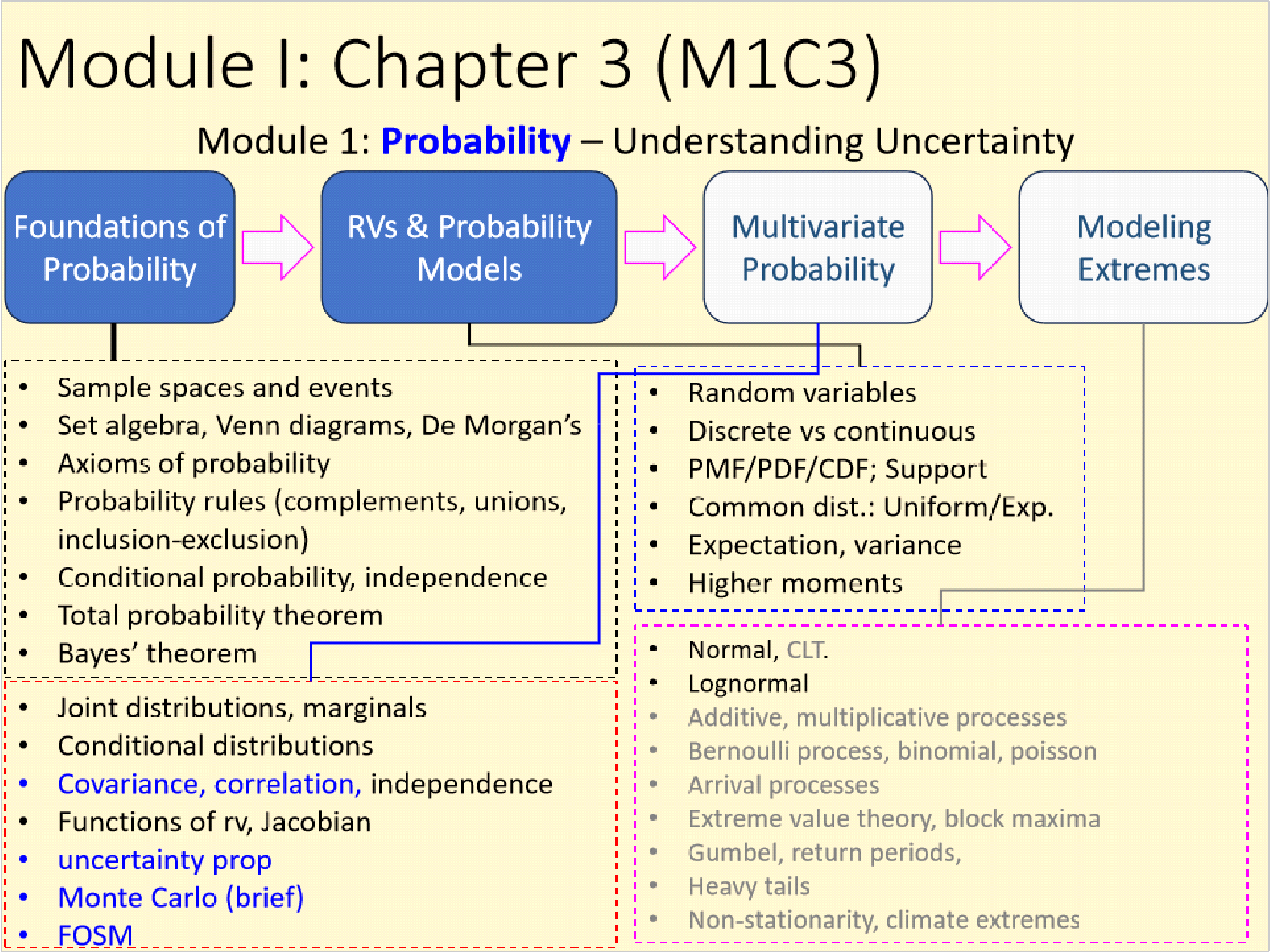
CV2090
Probability & Statistics for Civil Engineers
(Jul – Nov 2026)

Dr. Prakash S Badal

Joint Probability Distributions

Ang and Tang (Section 3.3)


Course Flow
Module 1: Probability – Understanding Uncertainty
Foundations of Probability
RVs & Probability Models
Multivariate Probability
Modeling Extremes
Module 2: Statistics – Learning from Data
Descriptive statistics
Statistical Inference
Regression & Statistical Learning
Module 3: Reliability – Decision under Uncertainty
Reliability Analysis
Computational Reliability
Stochastic Processes & Time-Dependent Reliability
Risk-Informed Decision Making


Module I: Chapter 3 (M1C3)
Module 1: Probability – Understanding Uncertainty
Foundations of Probability
RVs & Probability Models
Multivariate Probability
Modeling Extremes
Sample spaces and events
Set algebra, Venn diagrams, De Morgan's
Axioms of probability
Probability rules (complements, unions, inclusion-exclusion)
Conditional probability, independence
Total probability theorem
Bayes' theorem
Random variables
Discrete vs continuous
PMF/PDF/CDF; Support
Common dist.: Uniform/Exp.
Expectation, variance
Higher moments
Joint distributions, marginals
Conditional distributions
Covariance, correlation, independence
Functions of rv, Jacobian
uncertainty prop
Monte Carlo (brief)
FOSM
Normal, CLT.
Lognormal
Additive, multiplicative processes
Bernoulli process, binomial, poisson
Arrival processes
Extreme value theory, block maxima
Gumbel, return periods,
Heavy tails
Non-stationarity, climate extremes

L19

Sep 7, 2026

Messages from the Last Class

- **Message 1:** Joint RV

① Joint PDF

$$f_{XY}(x,y)$$

$$F_{XY}(x,y) = Pr(X \leq x, Y \leq y) = \int_{-\infty}^{x}\int_{-\infty}^{y} f_{XY}(u,v)\, du\, dv$$

② Joint PDF $\longrightarrow$ Marginal PDFs

$$\int_{-\infty}^{\infty} f_{XY}(u,y)\, du\, dy \longrightarrow f_Y(y)$$

Joint PDF $\longrightarrow$ conditional PDFs

$$f_{Y|X}(y|X=x) = \frac{f_{XY}(x,y)}{f_X(x)}$$

Joint probability distributions

P8. Independence

$$f_{XY}(x,y) \neq f_X(x)\, f_Y(y) \quad \text{in general}$$

If X & Y are statistically Independent

$$f_{XY}(x,y) = f_X(x) \cdot f_Y(y)$$

P9. Conditional

$$f_{Y|X}(y \mid X=x) = \frac{f_{XY}(x,y)}{f_X(x)}$$

$$\equiv f_{Y|X}(y|x)$$

$$E[g(x)] = \int_{-\infty}^{\infty} g(x) \cdot f_X(x)\, dx$$

P10. Expected value:

$g(x,y)$

$$E[g(x,y)] = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} g(x,y)\, f_{XY}(x,y)\, dx\, dy$$

7

Practice Problem

When $X \sim \mathcal{N}(\mu_X, \sigma_X^2)$ and $Y \sim \mathcal{N}(\mu_Y, \sigma_Y^2)$, their joint PDF is given by:

$$f_{XY}(x, y) =$$

$$\frac{1}{2\pi\sigma_X\sigma_y\sqrt{1-\rho^2}} \exp\left\{-\frac{1}{2(1-\rho^2)}\left[\left(\frac{x-\mu_X}{\sigma_X}\right)^2 - 2\rho\left(\frac{x-\mu_X}{\sigma_X}\right)\left(\frac{y-\mu_Y}{\sigma_Y}\right) + \left(\frac{y-\mu_Y}{\sigma_Y}\right)^2\right]\right\}$$

$$-\infty < x, y < \infty; \sigma_X, \sigma_Y > 0; -1 < \rho < 1$$

where ρ is the correlation coefficient between X and Y.

Find conditional PDF of Y given $X = x$.

$$f_X(x) = \frac{1}{\sqrt{2\pi}\,\sigma_X} \exp\left\{-\frac{1}{2}\left(\frac{x-\mu_X}{\sigma_X}\right)^2\right\}$$

→ PDF for $X \sim \mathcal{N}(\mu_X, \sigma_X^2)$

Conditional PDF $f_{Y|X}(y|x) = \dfrac{f_{XY}(x,y)}{f_X(x)}$

$$\int_{-\infty}^{\infty} e^{-z^2/2}\, dz = \sqrt{2\pi}$$

Joint probability distributions

P11. Moments

$$M_{nk} = E[X^n Y^k] = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} x^n y^k f_{XY}(x,y)\,dx\,dy$$

$$M_{00} = 1$$

$$M_{10} = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} x f_{XY}(x,y)\,dx\,dy = \int_{-\infty}^{\infty} x f_X(x)\,dx = \mu_X$$

$$M_{01} = \mu_Y$$

$$M_{11} = E[XY]$$

Joint probability distributions

P12. Central Moments

$$\eta_{nk} = E[(X-\mu_X)^n (Y-\mu_Y)^k]$$

$$\eta_{00} = 1$$

$$\eta_{10} = 0 \qquad \eta_{01} = 0$$

$$\eta_{20} = \sigma_X^2 \qquad \eta_{02} = \sigma_Y^2$$

$$\eta_{11} = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} (x-\mu_X)(y-\mu_Y)\, f_{XY}(x,y)\, dx\, dy$$

$$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} (xy f_{XY} - x\mu_Y f_{XY} - y\mu_X f_{XY} + \mu_X\mu_Y f_{XY})\, dx\, dy$$

$$= M_{11} - \mu_X\mu_Y - \mu_X\mu_Y + \mu_X\mu_Y$$

$$= E[XY] - \mu_X\mu_Y = E[XY] - E[X]E[Y]$$

P12a. COVARIANCE, $COV(X,Y) = E[XY] - E[X]E[Y]$

No squared σ_{XY} $= E[(X-\mu_X)(Y-\mu_Y)]$

Joint probability distributions

P13. Correlation Coefficient

$$\rho_{XY} = \frac{COV(X,Y)}{\sigma_X \sigma_Y} \equiv \frac{\sigma_{XY}}{\sigma_X \sigma_Y}$$

$$\Rightarrow COV(X,Y) \equiv \sigma_{XY} = \rho_{XY}\sigma_X\sigma_Y$$

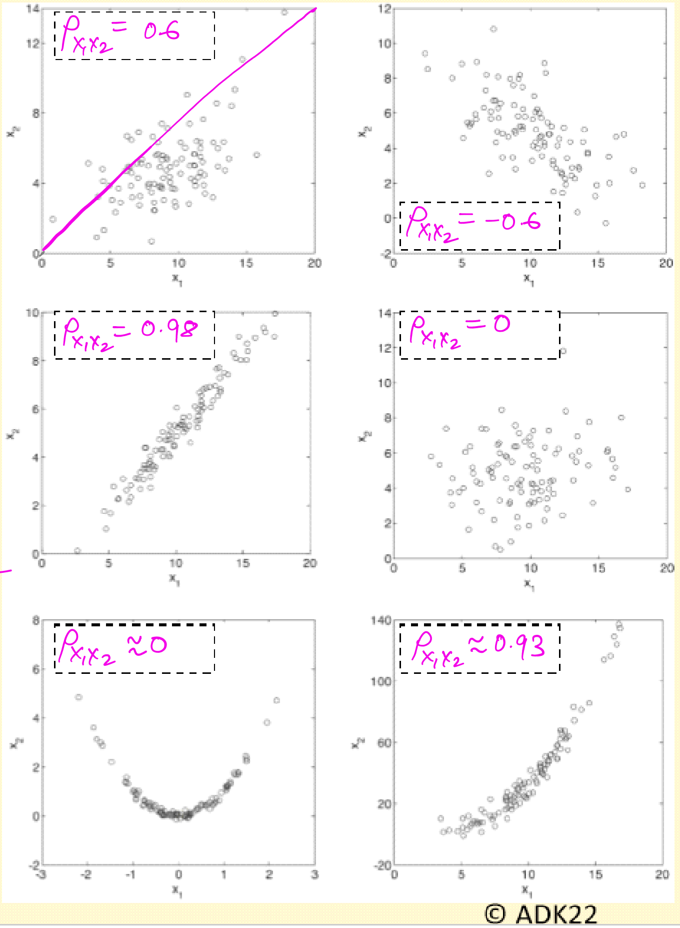
Correlation coefficient is a metric of predictability of "experience of one RV" given the "experience of another RV".

$\frac{X-\mu_X}{\sigma_X}$	$\frac{Y-\mu_Y}{\sigma_Y}$	
+1 −0.5 −2.0	+1 −0.5 −2.0	perfectly correlated $\rho_{XY} \to 1$
+1 −0.5 −2.0	−1 +0.5 +2.0	$\rho_{XY} \to -1$
+1 −0.5 −2.0	+1, −1 +0.5, −0.5 +2.0, −2.0	$\rho_{XY} \to 0$

Correlation coefficient

ρ measures linear dependence.

$\rho = 0 \not\Rightarrow$ statistically Independence



12

© ADK22

L20

Sep 8, 2026

13

Joint probability distributions

P14. Special case, Y is an affine transformation of X

$$Y = aX + b$$

$X: \mu_X, \sigma_X \qquad Y: \mu_Y, \sigma_Y$

What is COV[X, Y] and ρ_{XY}?

$$COV[X,Y] = E[XY] - E[X]\,E[Y]$$

$$= E[X(aX+b)] - \mu_X\,(a\mu_X + b)$$

$$= a\,E[X^2] + \cancel{b\,E[X]} - a\mu_X^2 \cancel{- b\mu_X} = a\,\sigma_X^2$$

$$\sigma_Y^2 = a^2\sigma_X^2 \Rightarrow \sigma_Y = |a|\,\sigma_X$$

$$\rho_{XY} = \frac{COV[X,Y]}{\sigma_X\,\sigma_Y} = \frac{a\,\sigma_X^2}{\sigma_X\,|a|\,\sigma_X} = \pm 1 \begin{cases} +1, & a>0 \\ -1, & a<0 \end{cases}$$

13

Practice Problem

Total gravity load T on a floor is the sum of dead load D and live load L

$$T = D + L$$

Derive the general formula for the variance of T, i.e., σ_T^2 in terms of σ_D^2, σ_L^2, and σ_{DL}.

$$\sigma_T^2 = E[T^2] - (E[T])^2$$

$$= \sigma_D^2 + \sigma_L^2 + 2\rho\sigma_D\sigma_L$$

$$Y = aX_1 + bX_2$$

$$\sigma_Y^2 = E[(Y-\mu_Y)^2] = E[\{(aX_1 + bX_2) - (a\mu_1 + b\mu_2)\}^2]$$

$$E[\{a(X_1-\mu_1) + b(X_2-\mu_2)\}^2]$$

$$= a^2 E[(X_1-\mu_1)^2] + b^2 E[(X_2-\mu_2)^2]$$

$$= a^2\sigma_X^2 + b^2\sigma_Y^2 + 2ab\rho\sigma_X\sigma_Y + 2ab\, E[(X_1-\mu_1)(X_2-\mu_2)]$$

Example on covariance

If X is either 11 or 17;   Y is either 20 or 8.

Find $\mathrm{COV[X, Y]}$ and ρ_{XY}.

$$S = \{(11,20), (11,8), (17,20), (17,8)\}$$

(a) (11,20) and (17,8) have 0.5 probability each.

(b) All four with 0.25 probability each.

$\mu_X = 14$
$\mu_Y = 14$

(a) $\mathrm{COV}[X,Y] = E[(X-\mu_X)(Y-\mu_Y)]$

$= \frac{1}{2}(-3 \times 6) + \frac{1}{2}(3 \times (-6)) = -18$

$\sigma_X = 3$ $\sigma_Y = 6$ $\rho_{XY} = -18/(3\times 6) = -1$

(b) $\mathrm{COV}[X,Y] = \frac{1}{4}(\cdots) + \frac{1}{4}(\cdots) + \frac{1}{4}(\cdots) + \frac{1}{4}(\cdots) = 0$

$\rho_{XY} = 0$

15

Mutual Fund: Diversification reduces *risk*

Mutual funds A and B, with equal average return, μ

Portfolio 1, P1: All investment in one of the funds, say, A $\rightarrow \mu, \sigma_A$

Portfolio 2, P2: w fraction in A and $(1-w)$ fraction in B $\rightarrow \mu, \sigma_B$

$\rho \equiv \rho_{A,B}$

Which one is preferable? Why?

R_{P2} : Random variable for return in Portfolio P2.

$R_{P2} = \omega \times R_A + (1-\omega) \times R_B$

$$E[R_{P2}] = \omega\mu + (1-\omega)\mu = \mu$$

$$\sigma^2[R_{P2}] = \omega^2\sigma_A^2 + (1-\omega)^2\sigma_B^2 + 2\omega(1-\omega)\rho\sigma_A\sigma_B$$

$$\sigma^2[R_{P2}] < \sigma_A^2$$

If $\rho = 1 \Rightarrow \sigma^2[R_{P2}] = \sigma_A^2 = \sigma_B^2$

If $\rho < 1$

Special Scenario $\omega = 1/2$, $\sigma_A = \sigma_B = \sigma$ $\Rightarrow \sigma_{R_{P2}} = \sigma\sqrt{\frac{1+\rho}{2}}$

Markowitz Model

Utility Theory (1940's)

Prospect Theory (1970's)

	$\sigma_{R_{P2}}$
$\rho = 1$	σ
$\rho = 0$	$\sigma/\sqrt{2}$
$\rho = -1$	0

M1C3 lec-2

Exact Uncertainty Propagation

Given the distribution of the inputs, how to determine the uncertainty of the output exactly?

Engineering equations → Functions of multiple RVs → Linear transformations → Nonlinear transformations → Jacobian → Exact analytical transformation → Monte Carlo (briefly, as *numerical exact* method)

Message: Engineering never measures outputs directly.

it propagates uncertainty through models.

Starting of notes already discussed in L13 (Aug 21, 2026)

17

Functions of Single rv

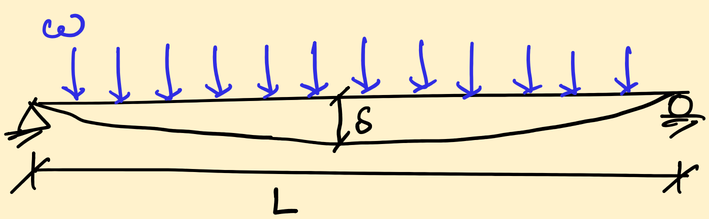
Ang A.H.S. and Tang (Section 4.2.1)

Why do we need functions of RV?

* wind speed $P_d = \frac{1}{2}\rho V^2$

$$V \longrightarrow P_d$$

* Beam



$$\delta = \frac{\omega L^4}{384 EI}$$

* Energy released in an Earthquake of magnitude M,

$$E = 10^{1.5M}$$

$E_7/E_6 \approx 32$

8

Why do we need functions of RV?

→ functions of 'observed RV' have physical significance.

$g(x) = 2x + 3$

$E[g(X)] = 2 \times E[X] + 3$

$\sigma^2[g(X)] = 2^2 \times \sigma_X^2$

$\Rightarrow \sigma[g(X)] = 2 \times \sigma_X$

$g(x) = x^2$

$E[g(x)] \stackrel{?}{=} g[E(x)]$

$g(x) = x^3 + 5\sin x + e^x$

$E[g(x)] = E[x^3] + 5E[\sin x] + E[e^x]$

Expectation of Functions

Expectation is a linear operator

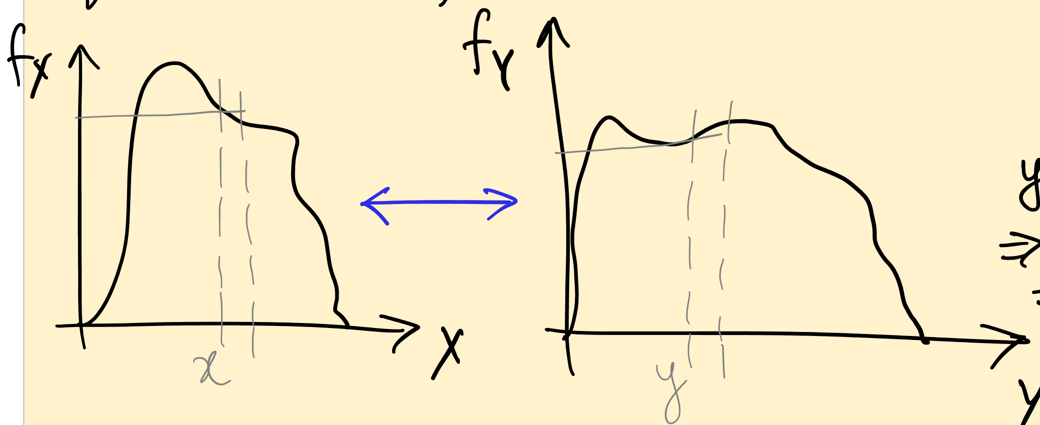
E & g(X) are commutative, if g(X) is an affine transformation $ax+b$

$$E[g(X)] = g[E(X)]$$

Function of Single RV

Let's say, X has a pdf $f_X(x)$

find PDF for $Y = X^3$.



$y = x^3$

$\Rightarrow x = y^{1/3}$

$\Rightarrow \frac{dx}{dy} = \frac{1}{3y^{2/3}}$

$f_X \, dx = f_Y \, dy$

$$\Rightarrow f_Y = f_X \left| \frac{dx}{dy} \right| = f_X(y^{1/3}) \cdot \left| \frac{1}{3y^{2/3}} \right|$$

Jacobian

M1C3 part-3/3
Approximate Uncertainty Propagation

What if the exact propagation is too complicated or impossible?

Exact propagation too difficult → Black-box engineering models → Taylor approximation → FOSM → Accuracy of approximation → Comparison with Monte Carlo

Message: Exactness is often too expensive.

Engineering relies on intelligent approximations.

Sources of uncertainties
Skip the sources example!

Sources of Uncertainty

- Inherent uncertainty: concrete strength; irreducible
- Statistical uncertainty: due to the lack of data; reducible
- Characteristics of system: material properties, dimensions
- Demand placed on the system
- Mathematical models used to analyze structure's behavior
- Measurements uncertainty: while assessing structure's health
- Probabilistic models used to describe uncertain quantities
- Human error

21

Categories of Uncertainty

In classical statistics,

- Aleatory

- Epistemic